

torch.nn.functional.poisson_nll_loss#

https://pytorch.org/docs/stable/generated/torch.nn.functional.poisson_nll_loss

Description:

Compute the Poisson negative log likelihood loss. See `PoissonNLLLoss` for details. `input` (Tensor) – Expectation of underlying Poisson distribution. `target` (Tensor) – Random sample $\text{target} \sim \text{Poisson}(\text{input})$. `log_input` (bool) – If True the loss is computed as $\exp(\text{input}) - \text{target} * \text{input} \exp(\text{input}) - \text{target} * \text{input}$, if False then loss is $\text{input} - \text{target} * \log(\text{input} + \text{eps})$ - $\text{target} * \log(\text{input} + \text{eps})$. Default: True

Function Signature

```
torch.nn.functional.poisson_nll_loss(input, target,
log_input=True, full=False, size_average=None, eps=1e-08,
reduce=None, reduction='mean')[source]#
```

Parameters

Return type

Returns:

Tensor

Detailed Documentation

Documentation

Compute the Poisson negative log likelihood loss.

See `PoissonNLLLoss` for details.

`input` (Tensor) – Expectation of underlying Poisson distribution.

`target` (Tensor) – Random sample $target \sim \text{Poisson}(input)$

`log_input` (bool) – If True the loss is computed as $\exp(input) - target * \exp(input)$, if False then loss is $input - target * \log(input + \epsilon)$. Default: True

`full` (bool) – Whether to compute full loss, i. e. to add the Stirling approximation term. Default: False

$$target * \log(target) - target + 0.5 * \log(2 * \pi * target) - \log(target) + 0.5 * \log(2 * \pi * target)$$

`size_average` (bool, optional) – Deprecated (see reduction).

`eps` (float, optional) – Small value to avoid evaluation of $\log(0)$ when

`log_input=False`. Default: `1e-8`

`reduce (bool, optional)` – Deprecated (see `reduction`).

`reduction (str, optional)` – Specifies the reduction to apply to the output: `'none'` | `'mean'` | `'sum'`. `'none'`: no reduction will be applied, `'mean'`: the sum of the output will be divided by the number of elements in the output, `'sum'`: the output will be summed. Note: `size_average` and `reduce` are in the process of being deprecated, and in the meantime, specifying either of those two args will override `reduction`. Default: `'mean'`